

FX Equity Spillover (Working Draft)

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1 Abstract

The aim of the model is to study and develop a strategy to trade foreign exchange (FX) based on equity returns. There is a well-known relationship between foreign equity returns and their respective currencies. While this relationship has been discussed in the literature, this work does not focus on the economic interpretation or wider economic consequences of the relationship. Instead, it explores whether the relationship can be modeled and exploited as a source of trading returns¹.

2 Layout of this Paper

This paper presents the strategy, model structure, empirical results, and supplementary theoretical work underlying the approach. Greater emphasis is placed on the backtest results of the signal and on rigorously evaluating their robustness from an econometric perspective. The first sections contain historical data and mathematical background to the strategy. The following sections contain the factor analysis, backtest results and optimized signal.

As the research and code base have been developed, the signal has been extended beyond momentum to include other equity factors. Additional extensions include the use of machine learning techniques and the extension of the analysis to single-name securities. While the code base and empirical testing is fairly rigorous, some of the more recent models show signs of overfitting and some results are likely too strong to be considered fully reliable at this stage.

The approach however offers a high level of fidelity in both its implementation and economic interpretation, especially given the breadth of successful results seen across the linear models and ETF-based strategies. Several potential sources of overstatement have been identified, including exchange-hour data snooping, overfitting in the machine learning models, and reliance on returns from relatively thinly traded FX markets. These issues are currently being addressed as the research continues. For transparency, all results are presented in this paper, while sections that remain under active development are explicitly labeled as “Working.”

3 Historical Data

The first model was built using front-month roll adjustments and FX futures that were acquired via the Bloomberg Terminal. FX futures offer adequate cross-sectional coverage for the models examined in this research, despite not accurately representing the larger FX market. Because they offer a practical tool for both historical backtesting and actual trading execution, FX futures were chosen. FX futures are beneficial from a backtesting standpoint since they simplify data handling, especially for non-deliverable forward

¹The full codebase and results are available at <https://github.com/diegodalvarez/FXEquitySpillover>

currencies, by embedding the carry involved in holding the underlying foreign currency.² From an implementation perspective, FX futures offer excellent historical data, are easily financed, and have comparatively high liquidity for major currency pairs.

Liquidity remains a consideration for some FX futures, as certain contracts exhibit relatively low trading volume. All currency futures included in the analysis are offered by CME, which generally provides reasonable liquidity across the contracts considered. Bloomberg total return series are also collected, as Bloomberg provides a broader set of FX return series than those available through futures alone. The cross-correlation between matched FX futures and Bloomberg Carry tickers is generally high. Asian currencies tend to exhibit lower cross-correlations between FX futures and Bloomberg Carry, which may be attributable to differences in index construction. However, these currencies also have the least amount of available data among the Bloomberg Carry indices, making it difficult to distinguish between differences arising from index construction and differences arising from the underlying data.

Since liquidity cannot be directly determined for the Bloomberg FX Carry indices due to the absence of a centralized market, the analysis instead separates these series into G10 and emerging-market currencies. The following table shows the available dates for the FX futures and Bloomberg Carry series.

Future	Name	Liquidity	Start Date	End Date
AD1 Curncy	Australian Dollar Future	Liquid	1987-01-13	2026-01-12
BP1 Curncy	British Pound Future	Liquid	1986-05-28	2026-01-12
BR1 Curncy	Brazilian Real Future	Liquid	1995-11-09	2026-01-12
CD1 Curncy	Canadian Dollar Future	Liquid	1986-04-04	2026-01-12
EC1 Curncy	Euro Foreign Future	Liquid	1998-05-20	2026-01-12
JY1 Curncy	Japanese Yen Future	Liquid	1986-05-23	2026-01-12
KO1 Curncy	Korean Won Future	Illiquid	2006-09-19	2026-01-12
NO1 Curncy	Norwegian Krone Future	Illiquid	2002-05-17	2026-01-12
NV1 Curncy	New Zealand Dollar Future	Liquid	1997-05-08	2026-01-12
PE1 Curncy	Mexican Peso Future	Liquid	1995-04-26	2026-01-12
PP1 Curncy	Polish Zloty Future	Illiquid	2004-07-12	2026-01-12
SE1 Curncy	Swedish Krona Future	Illiquid	2002-05-17	2026-01-12
SF1 Curncy	Swiss Franc Future	Liquid	1986-04-07	2026-01-12
SIR1 Curncy	Indian Ruppee Future	Illiquid	2013-01-29	2026-01-12
UCA1 Curncy	Offshore Chinese Renminbi Future	Illiquid	2012-09-18	2026-01-12

Table 1: FX Futures Historical Data

²For a detailed discussion of carry in FX futures, see Strategy Ten in *Advanced Futures Trading Systems* by Rob Carver.

Country	Index Name	Group	Start Date	End Date
Argentina	Argentine Peso Carry Return Index	EM Carry	1999-01-04	2026-03-31
Australia	Australian Dollar Carry Return Index	G10 Carry	1990-01-02	2026-03-31
Brazil	Brazilian Real Carry Return Index	EM Carry	1999-01-04	2026-03-31
UK	British Pound Carry Return Index	G10 Carry	1990-01-02	2026-03-31
Canada	Canadian Dollar Carry Return Index	G10 Carry	1990-01-02	2026-03-31
Chile	Chilean Peso Carry Return Index	EM Carry	1999-01-04	2026-03-31
China	Chinese Renminbi Carry Return Index	EM Carry	1998-12-14	2026-03-31
Colombia	Colombian Peso Carry Return Index	EM Carry	1999-01-04	2026-03-31
Denmark	Danish Krone Carry Return Index	EM Carry	1990-01-02	2026-03-31
Euro	Euro Carry Return Index	G10 Carry	1990-01-02	2026-03-31
Hong Kong	Hong Kong Dollar Carry Return Index	EM Carry	1990-01-02	2026-03-31
Iceland	Iceland Krona Carry Return Index	EM Carry	1999-01-04	2026-03-31
India	Indian Rupee Carry Return Index	EM Carry	1998-12-31	2026-03-30
Indonesia	Indonesian Rupiah Carry Return Index	EM Carry	1999-01-04	2026-03-31
Japan	Japanese Yen Carry Return Index	G10 Carry	1990-01-02	2026-03-31
Malaysia	Malaysian Ringgit Carry Return Index	EM Carry	1999-01-04	2026-03-31
Mexico	Mexican Peso Carry Return Index	EM Carry	1999-01-04	2026-03-31
New Zealand	New Zealand Dollar Carry Return Index	G10 Carry	1990-01-02	2026-03-31
Norway	Norwegian Krone Carry Return Index	G10 Carry	1990-01-02	2026-03-31
Peru	Peruvian Sol Carry Return Index	EM Carry	1999-01-04	2026-03-31
Philippines	Philippine Peso Carry Return Index	EM Carry	1999-01-04	2026-03-31
Poland	Polish Zloty Carry Return Index	EM Carry	1999-01-04	2026-03-31
Russia	Russian Ruble Carry Return Index	EM Carry	1999-01-04	2026-03-31
Singapore	Singapore Dollar Carry Return Index	EM Carry	1999-01-04	2026-03-31
South Africa	South African Rand Carry Return Index	EM Carry	1990-01-02	2026-03-31
Korea	South Korean Won Carry Return Index	EM Carry	1999-01-04	2026-03-31
Sweden	Swedish Krona Carry Return Index	EM Carry	1990-01-02	2026-03-31
Switzerland	Swiss Franc Carry Return Index	EM Carry	1990-01-02	2026-03-31
Taiwan	Taiwanese Dollar Carry Return Index	EM Carry	1999-01-04	2026-03-31
Thailand	Thai Baht Carry Return Index	EM Carry	1999-01-04	2026-03-31
Turkey	Turkish Lira Carry Return Index	EM Carry	1999-01-04	2026-03-31

Table 2: Bloomberg FX Carry Total Return Indices Historical Data

To model the equity component of the strategy, both ETFs and their corresponding equity benchmarks are used. The natural choice is to use VanEck ETFs and their respective benchmarks, which provide a consistent representation of the equity markets associated with each currency. Historical price data and other relevant attributes of the underlying indices are collected for each market. The table below presents the mapping between each country, its corresponding ETF, and the associated equity benchmark.

Country	ETF Ticker	Benchmark
Argentina	ARGT US Equity	M1AAR5R Index
Australia	EWA US Equity	NDDUAS Index
UK	EWU US Equity	NDDUUK Index
Brazil	EWZ US Equity	M1BR2550 Index
Canada	EWK US Equity	M1CXBLK Index
Chile	ECH US Equity	M1CL5IM Index
China	MCHI US Equity	M7CN Index
Colombia	COLO US Equity	M1CXGXB Index
Denmark	EDEN US Equity	M1DK5IM Index
Euro	EZU US Equity	NDDUEMU Index
Hong Kong	EWK US Equity	NU730026 Index
Iceland	GLCR US Equity	MVISLGTR Index
Israel	EIS US Equity	MISCNU Index
India	INDA US Equity	NDEUSIA Index
Indonesia	EIDO US Equity	NU724020 Index
Japan	EWJ US Equity	NDDUJN Index
Korea	EWY US Equity	M1KR2550 Index
Norway	ENOR US Equity	M1NO5IM Index
Malaysia	EWM US Equity	NDDUMAF Index
New Zealand	ENZL US Equity	NU756513 Index
Mexico	EWK US Equity	M1MX5IM Index
Poland	EPOL US Equity	M1PL5IM Index
Peru	EPU US Equity	M1PECAPD Index
Philippines	EPHE US Equity	NU710465 Index
Sweden	EWD US Equity	M1CXBISA Index
Russia	ERUS US Equity	MXRU500I Index
Singapore	EWS US Equity	M1CXBISC Index
South Africa	EZA US Equity	M1CXBAC Index
Switzerland	EWL US Equity	M1CH2550 Index
Taiwan	EWT US Equity	M1CXBISD Index
Thailand	THD US Equity	M1TH5IM Index
Turkey	TUR US Equity	NU724019 Index

Table 3: Country ETF and Benchmark Index Matching

Type	Fields
Equity	Cur. Mkt. Cap, Day to Day Ret. Gross Dividend, Fund Flow, Volume
Index	FCF (Est.), CUR EV/EBITDA (Est.), Div. Yield (Est.), EBIT (Est.), BEST_EPS, Gross Margin (Est.), P/E (Est.), P/Sales (Est.), Target Price (Est.), Cur. EV, EBIT, EBITDA, FCF Yield, Gros Margin, EPS, Net. Agg Div Yld, P/E Ratio, P/Sales

Table 4: Fundamental Fields by Type

The matched Bloomberg FX Carry Indices and FX futures exhibit high cross-correlations. However, the Bloomberg FX Carry indices are constructed in a long-dollar format, whereas FX futures are represented from the perspective of the underlying foreign currency. This difference in convention introduces an artificial negative correlation between the two series. To make the comparison consistent, returns are also volatility-targeted to a 10% annualized volatility target, using both lagged and contemporaneous hedge returns. The table below presents the correlations between the matched FX futures and Bloomberg FX Carry indices.

Country	Lagged	Contemporaneous	None
Australia	-0.90	-0.93	-0.91
Brazil	-0.68	-0.84	-0.84
Canada	-0.91	-0.92	-0.91
China	0.69	0.65	0.64
Euro	-0.95	-0.95	-0.95
India	-0.68	-0.67	-0.65
Japan	-0.95	-0.95	-0.94
Korea	-0.65	-0.66	-0.70
Mexico	-0.90	-0.91	-0.89
New Zealand	-0.89	-0.89	-0.89
Norway	-0.93	-0.93	-0.93
Poland	-0.95	-0.95	-0.95
Sweden	-0.93	-0.94	-0.91
Switzerland	-0.95	-0.94	-0.94
UK	-0.96	-0.96	-0.96

Table 5: Correlation between FX Futures and Bloomberg Carry Index

The correlations are quite close which is a good sign, and much of the lower correlation pairs come from a shorter history. The table below compares the number of days between FX Futures and Carry indices.

Country	Carry	Future
Australia	9456	9841
Brazil	7107	7597
Canada	9456	10044
China	7122	3334
Euro	9456	7007
India	7108	3266
Japan	9456	9995
Korea	7107	4825
Mexico	7107	7753
New Zealand	9456	7248
Norway	9456	5955
Poland	7107	5417
Sweden	9456	5959
Switzerland	9456	10026
UK	9456	9999

Table 6: Number of Days in Dataset

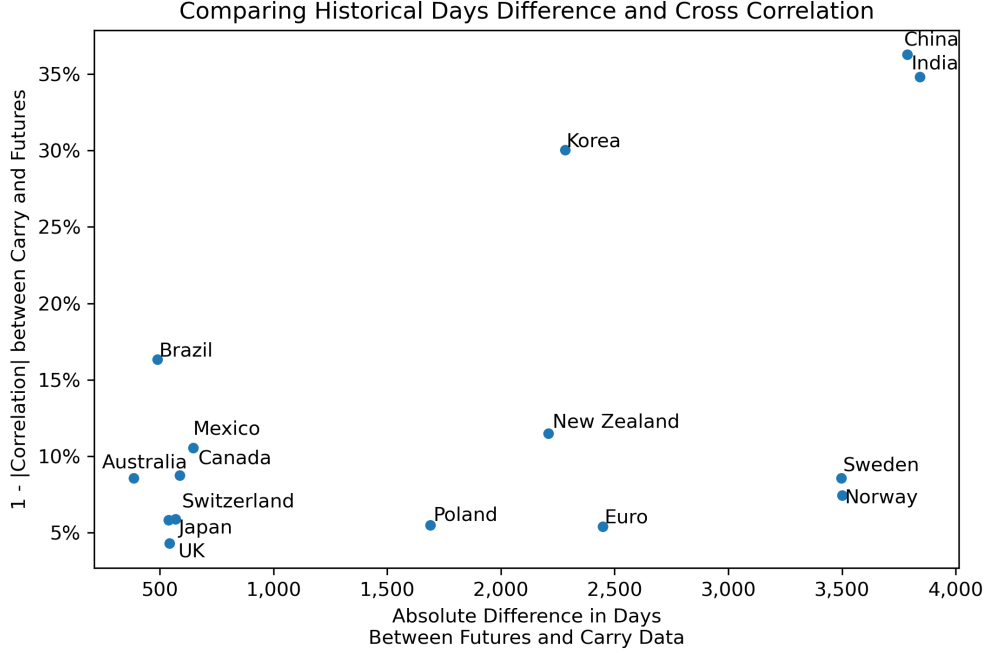


Figure 1: Correlation vs. Days Spread

4 Theoretical Approach

Following Cenedese et. al [2] the theoretical foundation around this model follows a no-arbitrage constraint. For a domestic and foreign equity return R_d and R_f where the foreign return has to be scaled by some currency s_t

$$\mathbb{E}(R_d \cdot m_h) = \mathbb{E}\left(R_f \cdot \frac{s_{t+1}}{s_t} \cdot m_h\right) = 1 \quad (1)$$

m_h is the discounting factor to maintain no-arbitrage also called the stochastic discount factor. Then rearrange the formula using covariance rules.

$$\mathbb{E}(R_d) \cdot \mathbb{E}(m_h) + \text{cov}(m_h, R_d) = \mathbb{E}\left(R_f \cdot \frac{s_{t+1}}{s_t}\right) \cdot \mathbb{E}(m_h) + \text{cov}\left(m_h, R_f \cdot \frac{s_{t+1}}{s_t}\right) = 1 \quad (2)$$

Then let $\mathbb{E}(m_h)$ be the one-period, risk free zero-coupon bond R_b in country d .

$$\mathbb{E}(R_d) \cdot \frac{1}{R_b} + \text{cov}(m_h, R_d) = \mathbb{E}\left(R_f \cdot \frac{s_{t+1}}{s_t}\right) \cdot \frac{1}{R_b} + \text{cov}\left(m_h, R_f \cdot \frac{s_{t+1}}{s_t}\right) = 1 \quad (3)$$

The equation above is useful since it separates the contribution of risk premia and the risk free rate. In a no-arbitrage setting, the stochastic discount factor m_h discounts the returns back to one. If the discounting is too slow - corresponding to positive excess returns - the covariance term $\text{cov}(m_h, R_d)$ adjusts to offset these returns back to one. Conversely, if discounting is too fast, implying negative excess returns, the covariance term adjusts in the opposite direction. Then re-arrange the formula to separate the equity returns (excess returns) and the risk premia

$$\underbrace{\mathbb{E}\left(R_f \cdot \frac{s_{t+1}}{s_t}\right) \frac{1}{R_b} - \mathbb{E}(R_d) \cdot \frac{1}{R_b}}_{\text{excess equity return differential}} = \underbrace{\text{cov}(m_h, R_d) - \text{cov}\left(m_h, R_f \cdot \frac{s_{t+1}}{s_t}\right)}_{\text{difference in risk premia}} \quad (4)$$

Now, to fully isolate the currency returns, use a similar approach for log returns. Therefore,

$$\mathbb{E}\left(\exp\left(\log\left(m_h R_f \frac{s_{t+1}}{s_t}\right)\right)\right) = \mathbb{E}(\exp(\log(m_h R_d))) \quad (5)$$

Equation 5 can be rewritten as

$$\mathbb{E}(\exp(\tilde{m} + r_f + \Delta s_{t+1})) = \mathbb{E}(\exp(\tilde{m} + r_d)) \quad (6)$$

Where r_d , r_f , and Δs_{t+1} are the log domestic and foreign equity with changes in the currency, $\tilde{m}$ is $\log(m_h)$. Then with some variance and covariance rules we can get the log-comparable to equation 3.

$$\mathbb{E}(\tilde{m}) + \mathbb{E}(r_f) + \mathbb{E}(\Delta s_{t+1}) + \frac{1}{2}\text{var}(\tilde{m} + r_f + \Delta s_t) = \mathbb{E}(\tilde{m}) + \mathbb{E}(r_d) + \frac{1}{2}\text{var}(\tilde{m} + r_d) \quad (7)$$

Then re-arrange

$$\mathbb{E}(r_f) + \mathbb{E}(\Delta s_{t+1}) - \mathbb{E}(r_d) = \frac{1}{2}(\text{var}(\tilde{m} + r_f + \Delta s_t) - \text{var}(\tilde{m} + r_d)) \quad (8)$$

$$= \frac{1}{2}(\text{var}(r_f + \Delta s_t) - \text{var}(r_d)) + \text{cov}(\tilde{m}, r_f + \Delta s_{t+1}) - \text{cov}(\tilde{m}, r_d) \quad (9)$$

Which results in an equation very similar to equation 3 but in log form

$$\mathbb{E}(\Delta s_{t+1}) = \underbrace{\mathbb{E}(r_d) - \mathbb{E}(r_f)}_{\text{expected log return differential}} + \underbrace{\text{cov}(\tilde{m}, r_f + \Delta s_{t+1}) - \text{cov}(\tilde{m}, r_d)}_{\text{difference in risk premia}} + \underbrace{\frac{1}{2}(\text{var}(r_f + \Delta s_t) - \text{var}(r_d))}_{\text{log-variance adjustment}} \quad (10)$$

The approach is appealing because it preserves flexibility in the sign of the covariance term $\text{cov}\left(m_h, R_j \frac{S_{t+1}}{S_t}\right)$. Hau and Rey [4] argue that portfolio rebalancing by international investors implies a negative relationship, whereas Bohn and Tesar[1], and Froot, Scharfstein, and Stein[3] document evidence consistent with a positive relationship.

Cenedese et al further this framework beyond equity returns and use other equity factors such as dividend yield, value, and momentum. Those other factors will be considered but require empirical analysis beforehand.

5 Examining the Factor Exposure

The overall objective is to analyze the FX exposure of the return differential between FX and equity markets. As a first step, sovereign equity exposure is accounted for to establish an appropriate benchmark. Both FX and Equity ETF returns are volatility-targeted to 10% annualized volatility to control for time-varying volatility. Volatility is estimated using an exponentially weighted standard deviation based on the previous 100 trading days. Both contemporaneous and lagged hedging specifications are considered to account for differences in the timing of the hedge.

Factor exposure is measured by the β coefficient from an OLS regression. The regressions are estimated contemporaneously rather than using lagged factor returns to capture the instantaneous relationship between the factor and the dependent variable. For brevity, the OLS specifications are condensed throughout this whitepaper given the large number of regressions and variables considered.³

The analysis initially uses descriptive statistics to summarize the results. p -values are not reported throughout the analysis because the majority are well below the 5% significance threshold; unless otherwise stated, statistical significance is therefore assumed at the 5% level.

We first compare the estimated β coefficients from regressions using volatility-targeted returns with those from regressions using unadjusted returns. The volatility adjustment can improve both the estimated factor exposure and the corresponding t -statistics. The table below reports the proportion of regressions for which each regression statistic improves following the volatility adjustment.

³The complete regression specifications, results, and tables are available in the `1FactorModel.ipynb` notebook.

Improved Param group	No Vol Target		With Vol Target	
	Coef	t -value	Coef	t -value
All	0.65	0.89	0.35	0.11
EM Carry	0.48	0.83	0.52	0.17
G10 Carry	0.86	1.00	0.14	0.00
Illiquid Futures	0.71	0.86	0.29	0.14
Liquid Futures	0.89	1.00	0.11	0.00

Table 7: Comparing proportion β and t -value improvements when using volatility targeting (contemporaneous) of OLS regression of FX Returns vs ETF returns

The proportions are relatively the same when comparing the returns differential with and without volatility targeting.

Improved Param group	No Vol Target		With Vol Target	
	Coef	t -value	Coef	t -value
All	0.65	0.89	0.35	0.11
EM Carry	0.48	0.83	0.52	0.17
G10 Carry	0.86	1.00	0.14	0.00
Illiquid Futures	0.71	0.86	0.29	0.14
Liquid Futures	0.89	1.00	0.11	0.00

Table 8: Comparing proportion β and t -value improvements when using volatility targeting (contemporaneous) of OLS regression of FX Returns vs ETF - SPY returns

We begin by taking the average β , t -statistic, and R^2 across regressions for the raw ETF returns, the ETF return differential, and the trend of the ETF return differential, defined as its 30-day exponentially weighted z -score. The table below summarizes the results for the reported regressions.

Group	Vol. Target Exog	None			Perf		
		Coef	R^2	t -value	Coef	R^2	t -value
EM Carry	ETF Returns	0.1539	0.2076	35.2131	0.1441	0.1838	31.9543
	... Net SPY	0.1758	0.1627	29.6584	0.1628	0.1491	27.8395
	... Trend	0.0025	0.1408	27.2044	0.0024	0.1493	28.1042
G10 Carry	ETF Returns	0.2241	0.2978	48.4235	0.1983	0.2463	41.4727
	... Net SPY	0.2699	0.2002	38.3286	0.2530	0.1832	36.0044
	... Trend	0.0030	0.1886	36.9780	0.0030	0.1959	37.7787
Illiquid Futures	ETF Returns	0.1917	0.3054	38.8232	0.1674	0.2667	34.8097
	... Net SPY	0.2462	0.2513	33.8082	0.2127	0.2254	31.1646
	... Trend	0.0030	0.2088	32.8322	0.0031	0.2262	34.7236
Liquid Futures	ETF Returns	0.2039	0.2580	46.7742	0.1697	0.2205	41.4517
	... Net SPY	0.2653	0.2114	42.3436	0.2336	0.1938	39.8616
	... Trend	0.0033	0.2006	40.8613	0.0030	0.2111	42.1721

Table 9: Comparing average OLS stats for various raw ETF returns, returns spread, and returns spread signal (ran separately) controlling for volatility targeted returns (hedged contemporaneously)

The model is also estimated using the lagged signal to evaluate its ability to predict subsequent futures returns. Although the initial β coefficients for the returns differential trend signal are relatively small, making

direct comparison between the lagged and contemporaneous specifications difficult, the overall quality of the regression decreases when using the lagged signal. The table below compares the regression results based on the average OLS statistics across independently estimated regressions.

Group	Lag Vol.	Stat Sig Rate		coeff		p-value		R^2		t-stat	
		No Lag	Lagged	No Lag	Lagged	No Lag	Lagged	No Lag	Lagged	No Lag	Lagged
EM	None	1.0000	0.6522	-0.0025	-0.0003	0.0000	0.1588	0.1408	0.0047	-27.2044	-3.3816
	Perf	1.0000	0.6087	-0.0024	-0.0003	0.0000	0.1700	0.1493	0.0051	-28.1042	-3.3929
G10	None	1.0000	0.2857	-0.0030	0.0001	0.0000	0.2891	0.1886	0.0003	-36.9780	0.7586
	Perf	1.0000	0.2857	-0.0030	0.0001	0.0000	0.3249	0.1959	0.0004	-37.7787	0.9198
Ill Fut	None	1.0000	0.7143	0.0029	0.0002	0.0000	0.1263	0.2088	0.0025	29.8152	1.1120
	Perf	1.0000	0.7143	0.0027	0.0001	0.0000	0.1081	0.2262	0.0026	31.6258	1.2801
Liq Fut	None	1.0000	0.2222	0.0033	0.0001	0.0000	0.3883	0.2006	0.0010	40.8613	1.1114
	Perf	1.0000	0.2222	0.0030	0.0001	0.0000	0.4221	0.2111	0.0009	42.1721	0.8596

Table 10: Comparison of contemporaneous and lagged OLS regression results across asset groups and volatility-targeting specifications.

6 Returns Momentum Model

This section will cover the results of trading the momentum model. To keep the portfolio optimization simple when results are reported they are done with equal weighting across FX carry or future. All returns are volatility targeted at the signal-return-level at a 10% annualized volatility level. Most of the plots use contemporaneously volatility-targeted returns. Lagged hedging is taken into consideration and shown in the tables.

The sign of the raw return differential and the return differential spread is first applied without imposing any prior assumption about their relationship, and the resulting signals are used to construct the corresponding backtest portfolios. The two portfolios exhibit a high degree of correlation, with a correlation of approximately $\sim 80\%$.

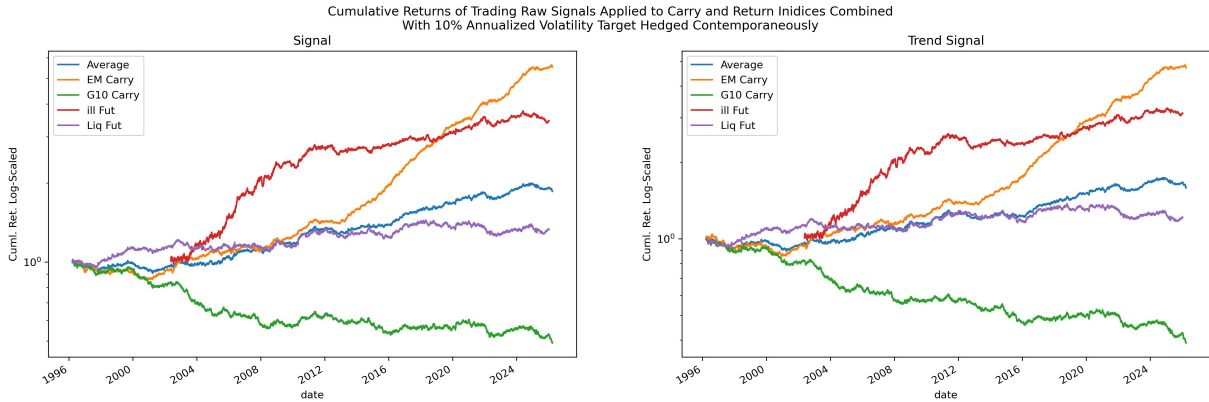


Figure 2: Backtest of applying returns differential and trend of returns differential to all asset classes with 10% annualized volatility hedged contemporaneously

Below is the Sharpe ratio performance of the portfolios, which is relatively strong given that the signals are applied without any model fitting.

Backtest	Group	Lagged	Perfect
Signal	Average	0.53	0.56
	EM Carry	1.43	1.45
	G10 Carry	-0.43	-0.43
	Liq Fut	0.21	0.24
	ill Fut	0.82	0.88
Trend Signal	Average	0.43	0.42
	EM Carry	1.29	1.31
	G10 Carry	-0.55	-0.58
	Liq Fut	0.18	0.17
	ill Fut	0.77	0.81

Table 11: Comparing sharpe of ETF returns and returns spread (applied agnostically) backtest controlling for volatility targeted returns

The analysis now turns to the forecasted OLS model. Before examining the backtest, the relationship between the OLS parameters and performance is first examined on a single-security basis. The plot below compares the β and t -statistic of the OLS model with the security's relative performance. The Sharpe ratio is calculated using volatility-hedged performance. The plot shows that when the OLS model detects a stronger relationship, as indicated by a larger β or t -statistic, the quality of the forecast tends to increase, resulting in a higher Sharpe ratio.

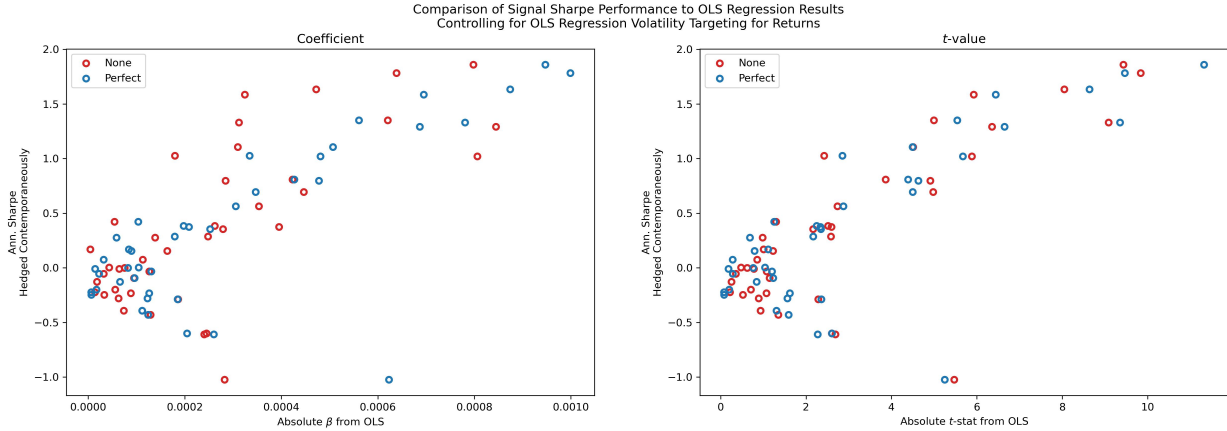


Figure 3: Relationship between OLS model parameters and security-level forecast performance. The figure compares the β and t -statistic of the OLS model with the corresponding security's relative performance.

The resulting backtest, with equal weighting applied after volatility targeting, produces the following PnL curves.

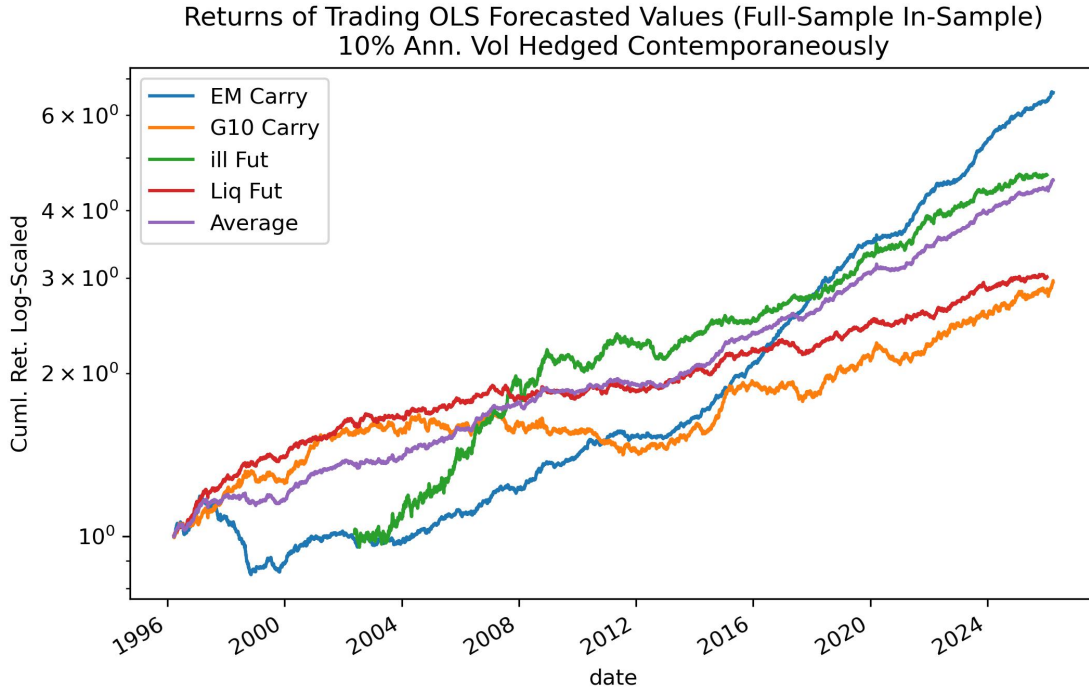


Figure 4: Cumulative PnL of the equal-weighted portfolios following volatility targeting.

Using 2018 as the cutoff period for the training dataset and an expanding out-of-sample approach the sharpes for the same backtests are in the table below.

Vol. Target Group	Expanding		Full Sample		Train/Test	
	Lagged	Perfect	Lagged	Perfect	Lagged	Perfect
Average	1.30	1.34	1.76	1.85	2.07	2.17
EM Carry	1.53	1.56	1.65	1.74	1.77	1.87
G10 Carry	0.22	0.25	0.72	0.76	0.84	0.89
Liq Fut	0.36	0.35	1.05	1.13	1.08	1.17
ill Fut	0.97	1.00	1.06	1.09	1.08	1.09

Table 12: Sharpes for OLS forecasted portfolios controlling for sample size.

The performance is quite strong even out of sample. The train/test approach leads to slightly higher Sharpe ratios, which implies that there is some time-varying component captured in the earlier part of the dataset. This should be noted as an attribute of the results, but given that the expanding out-of-sample OLS performance remains reasonable, it is not a major concern.

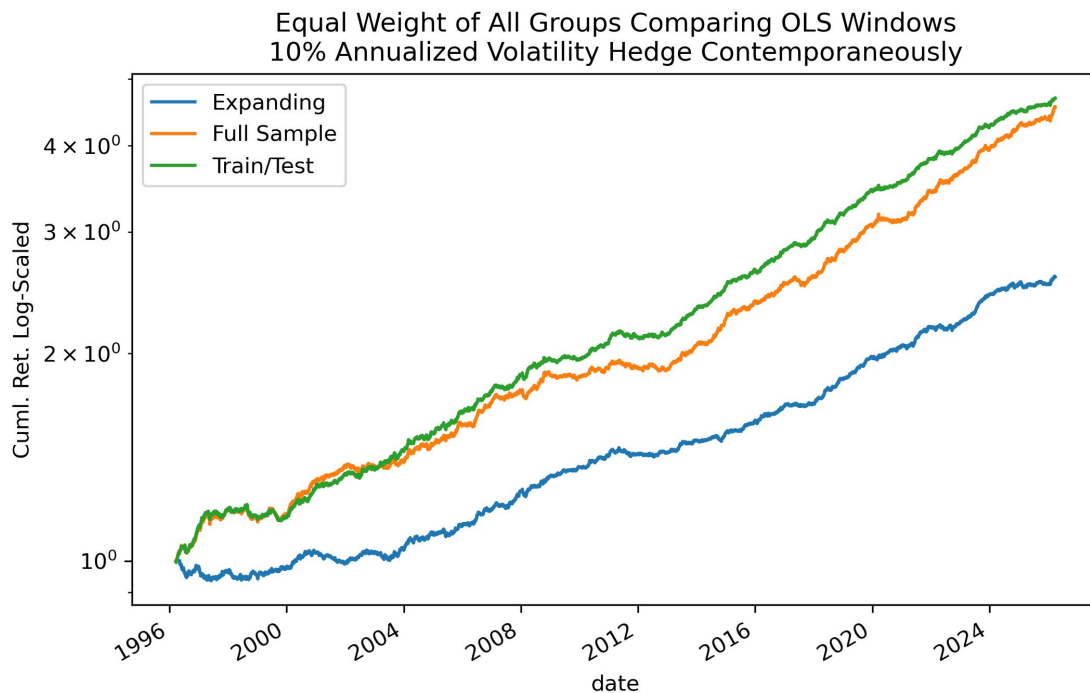


Figure 5: Cumulative PnL of the equal-weighted portfolios controlling for OLS Sample size using 2018 as cutoff for training data.

The groupings complement each other well, as they exhibit relatively low cross-correlation among themselves. Below is the correlation matrix of the expanding out-of-sample returns.

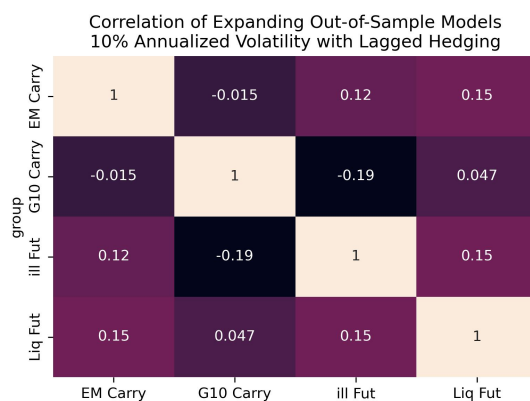


Figure 6: Correlation matrix of expanding out-of-sample returns across the different groupings.

An alternative strategy can be made by taking the instantaneous OLS regression, calculating the residuals, and shorting them, which is akin to calculating the fair value via OLS and taking a position to trade its reversion back to fair value. The performance is surprisingly weaker than the forecasted model, which was not initially implied by the OLS results.

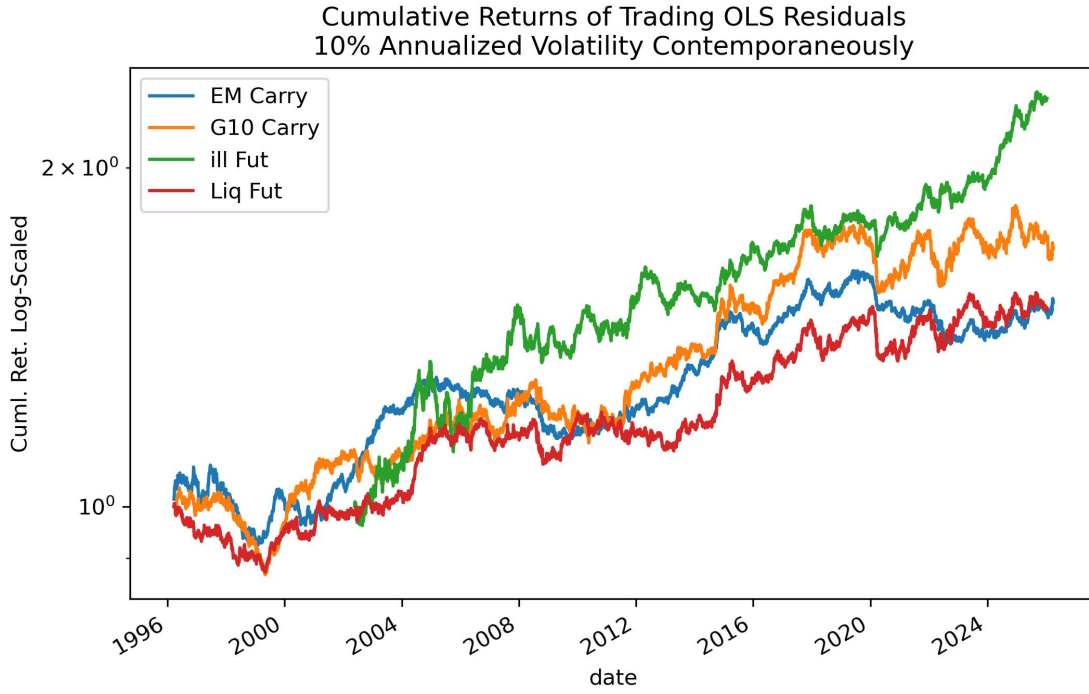


Figure 7: Backtest of *shorting* residuals of OLS model, full-sample/in-sample using 10% Annualized Volatility Hedged Contemporaneously.

The corresponding Sharpe ratios using the same train/test split approach are shown in the table below.

Vol. Target Group	Expanding		Full Sample		Train/Test	
	Lagged	Perfect	Lagged	Perfect	Lagged	Perfect
Average	0.53	0.60	0.48	0.52	0.46	0.50
EM Carry	0.68	0.78	0.32	0.37	0.47	0.54
G10 Carry	0.29	0.34	0.33	0.35	0.26	0.27
Liq Fut	0.21	0.24	0.30	0.31	0.20	0.22
ill Fut	0.54	0.60	0.54	0.59	0.61	0.65

Table 13: Sharpes of *shorting* residuals of OLS model, full-sample/in-sample using 10% Annualized Volatility.

The plot below compares the performance of the backtest.

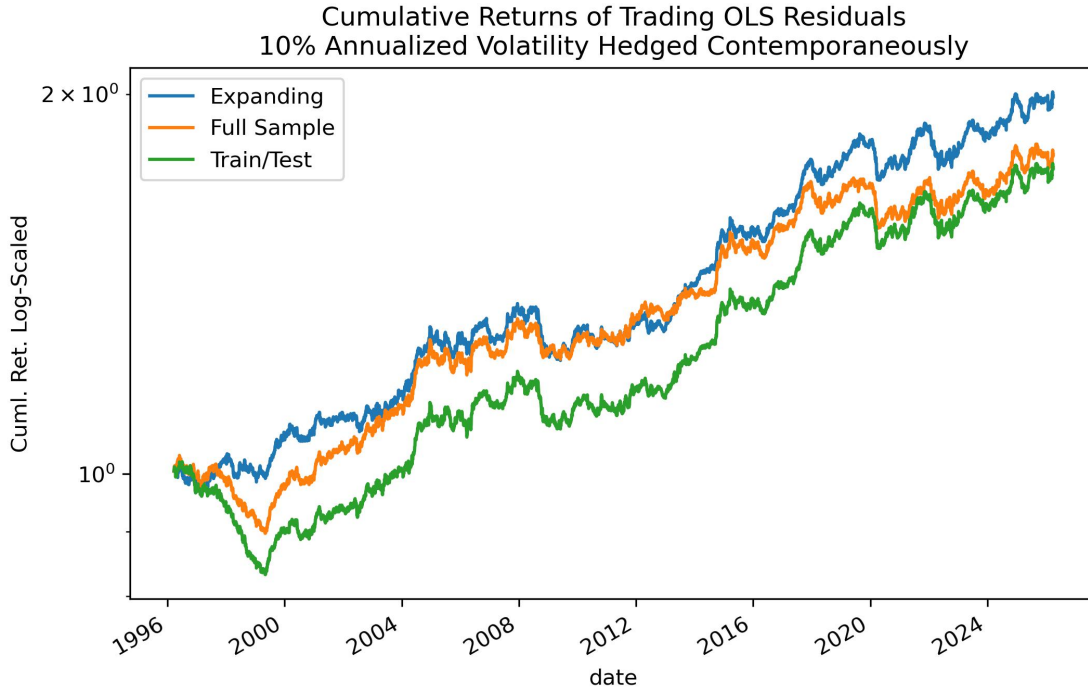


Figure 8: Backtest of *shorting* residuals of OLS model, controlling for sample size, with 10% Annualized Volatility Hedging.

A simple signal optimization can be performed. The approach is to take the signal and decile the values, then calculate the next-day Sharpe for each decile bin. The outer two deciles are then examined, and a position is taken corresponding to the Sharpe when the two deciles agree; otherwise, no trade is made. The middle deciles are not considered. This deciling approach helps identify non-linear patterns, avoid trading weak outliers, minimize transaction costs by reducing two-way trading to one-way trading, and concentrate trades at the most extreme signal values.

On the following page is the deciled approach for the returns differential trend signal. Since the values are already normalized via exponentially weighted z-score the deciles can be applied directly. Regarding the optimization and backtest, the optimization is done using contemporaneously volatility targeted returns while the backtest applies the signal to the raw returns and then volatility targets the signal returns.

Cumulative Returns of Trading Decided Optimized Returns
10% Annualized Volatility Target Hedged Contemporaneously

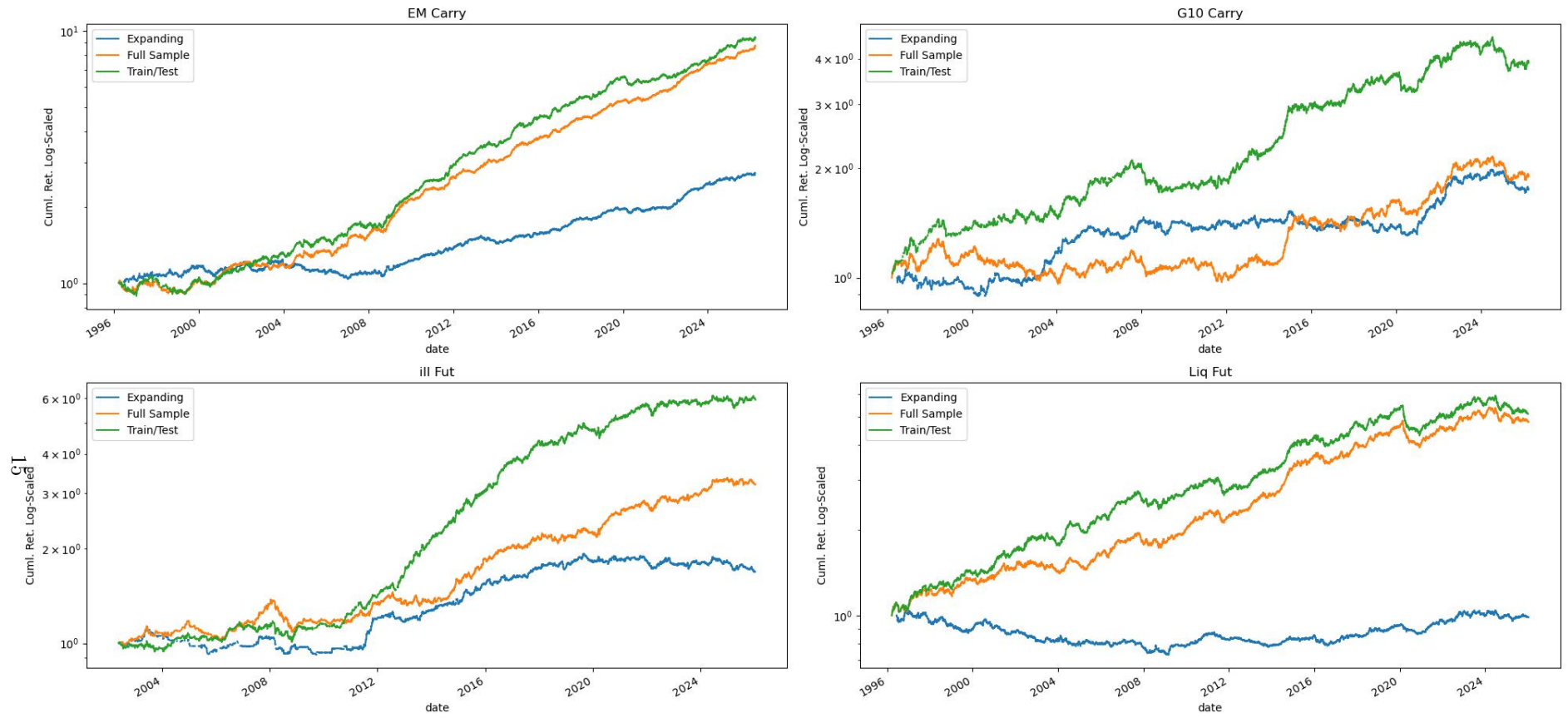


Figure 9: Backtest of decile optimized signal

Below is the table of sharpe ratios for the portfolios in the prior graph.

Vol. Target Group	Expanding		Full Sample		Train/Test	
	Lagged	Perfect	Lagged	Perfect	Lagged	Perfect
EM Carry	1.14	1.36	1.72	1.78	1.77	1.77
G10 Carry	0.49	0.54	0.83	0.84	0.73	0.75
Liq Fut	0.41	0.58	0.99	1.05	1.08	1.14
ill Fut	1.22	1.29	1.20	1.19	1.01	1.06

Table 14: Deciled optimized Signal backtest sharpes ratios

Although the sharpes are lower than the OLS forecasted, the lower cross-correlation ends up becoming a big difference maker. Below is a table of equal weight backtest across all the groups.

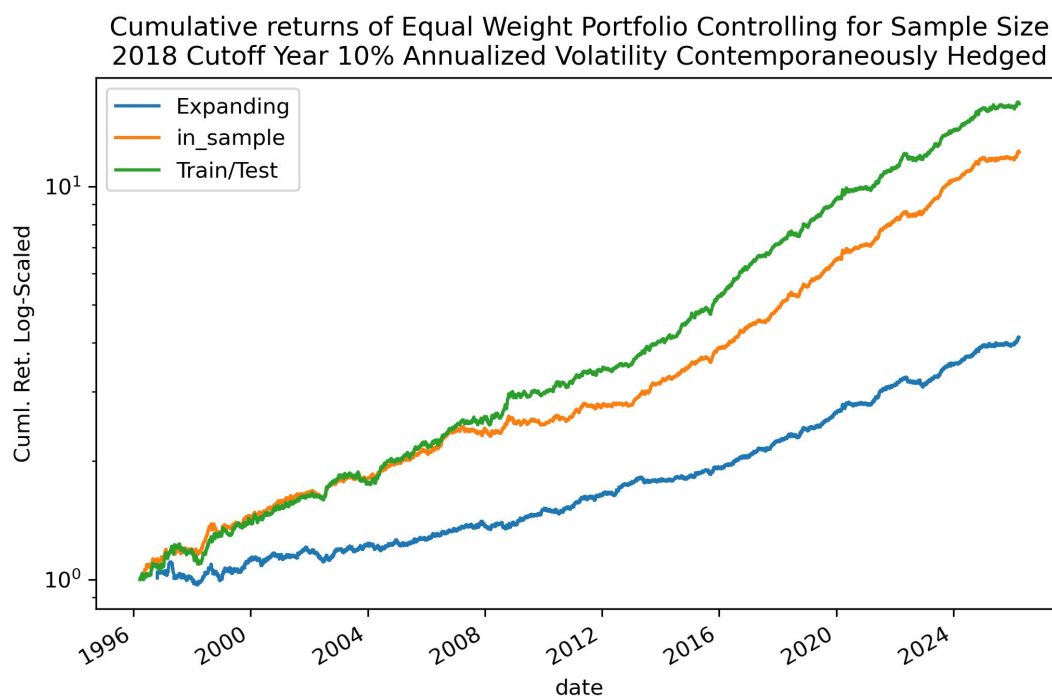


Figure 10: Backtest of Signal Decile Optimized Portfolio

Its corresponding sharpe ratios.

Optimization	Sharpe Ratios	
	Lagged	Perfect
Expanding	1.22	1.43
Train/Test	1.81	1.86
Full Sample	1.97	2.01

Table 15: Sharpe of Deciled Signal Backtest

The same deciling approach can also be applied to the OLS residuals. Following the same optimization procedure results in the following Sharpe ratios.

Vol. Target Group	Expanding		Full Sample		Train/Test	
	Lagged	Perfect	Lagged	Perfect	Lagged	Perfect
EM Carry	0.76	0.86	1.31	1.41	1.24	1.35
G10 Carry	0.38	0.43	0.38	0.37	0.76	0.82
Liq Fut	0.04	0.02	0.81	0.86	0.78	0.89
ill Fut	0.47	0.55	0.85	0.88	1.29	1.37

Table 16: Deciled optimized OLS Residual backtest sharpes ratios

The backtest of the stratgey can be seen the PnL curve.

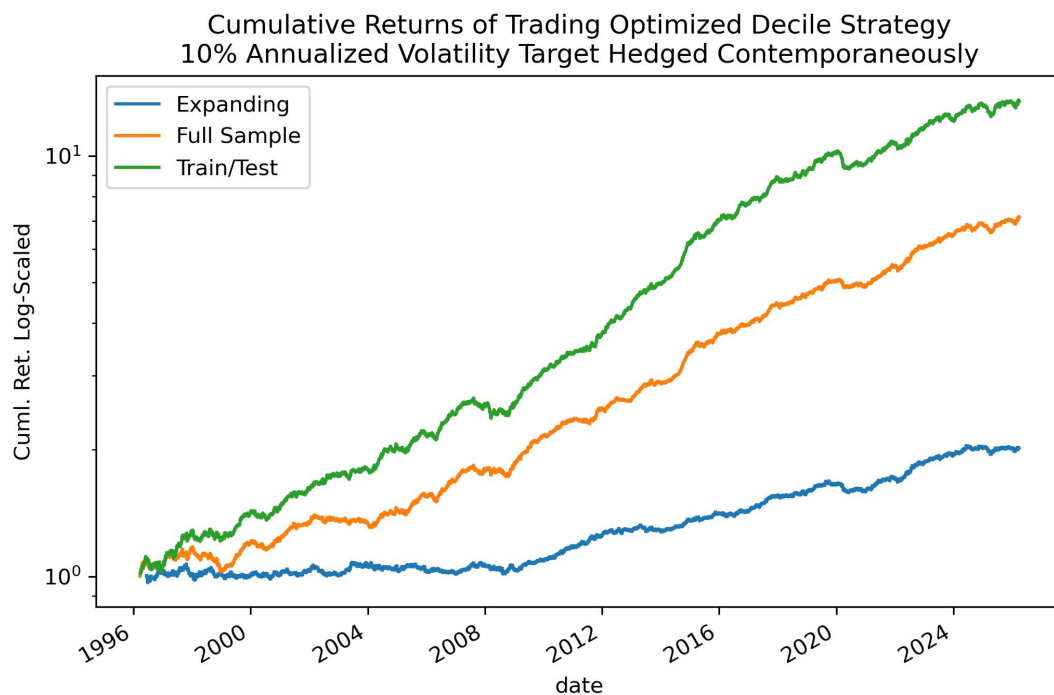


Figure 11: Backtest of OLS Residual Decile Optimized Portfolio controlling fos train size

The sharpes of those backtests can be seen below in the table.

Vol. Target	Lagged	Perfect
Expanding	0.74	0.79
Full Sample	1.41	1.48
Train/Test	1.52	1.66

When the portfolios are combined, they exhibit relatively low cross-correlation with each other. The bar chart below shows the average cross-correlation across the strategies, controlling for the training period.

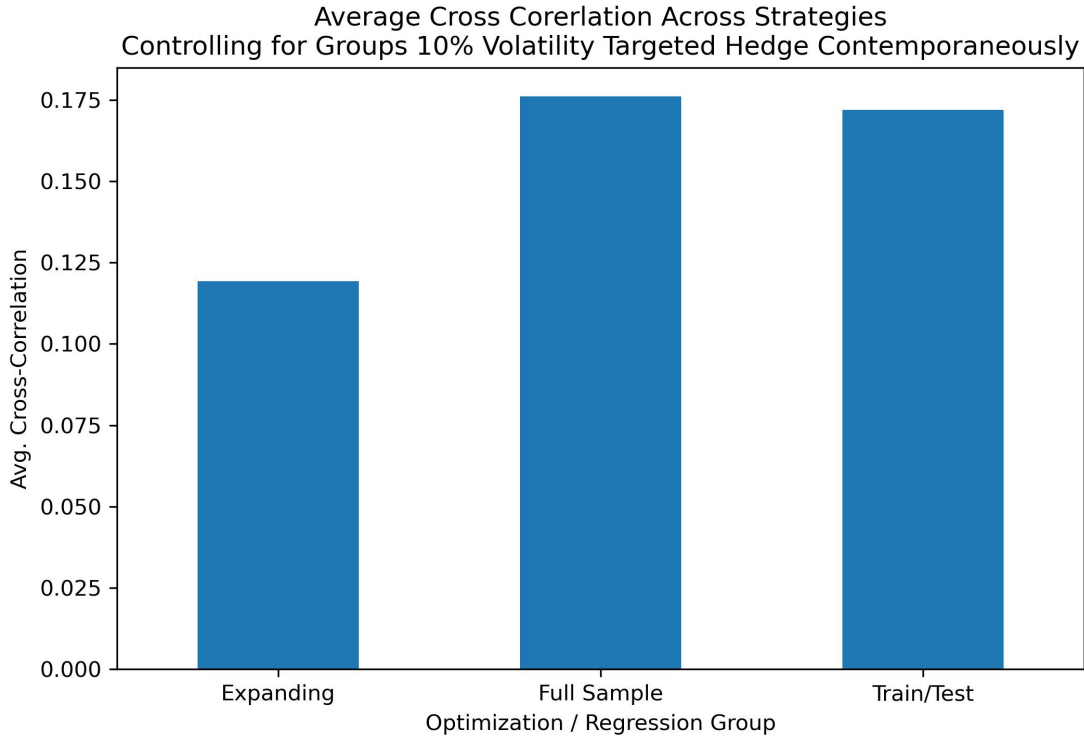


Figure 12: Average Cross Correlation amongst all presented strategies controlling for sample group

The sharpes for each of the portfolios as well as average between all can be seen in the table.

Regression	Average	OLS For	OLS Resid	Opt Resid	Opt Trend
Expanding	1.78	1.42	0.69	0.81	1.43
Full Sample	2.30	1.85	0.50	1.48	1.99
Train/Test	2.46	2.25	0.51	1.66	1.91

7 Expanding to the Factor Zoo

Following the work of Cenedese, Payne, and Valente[2], the analysis can be extended to other equity fundamentals. Using the factors listed in Table 4, the same approach can be applied. A small adjustment is made to the factors by taking their daily difference and dividing by their 10-day exponentially weighted standard deviation. The purpose of this transformation is to smooth out some of the jumps that occur as a result of changes in accounting data and analyst estimates.

Below is a plot showing the comparison of the adjusted fundamental data vs the raw fundamental data.

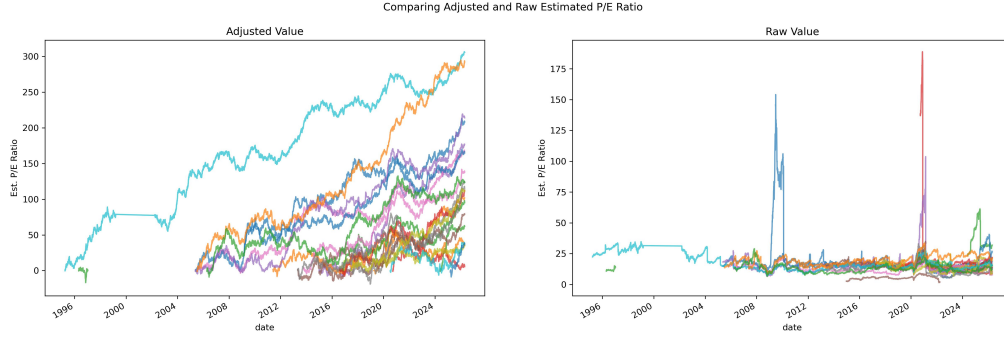


Figure 13: Comparing volatility adjusted factor data vs raw data

Due to the large number of regressions and resulting observations, descriptive statistics are used to summarize the regression results. The table below reports the average OLS parameters across the regressions, along with the number of statistically significant factors. The false positive rate and expected number of statistically significant factors are also considered to account for the possibility of finding spurious relationships by chance.

field	Coeff	p -value	R^2	t -value	Stat. Sig.	Expected	FDR Sig
Est FCF	0.0004	0.1863	0.0048	2.6748	18	1.7500	0.5143
Est EV/EBITDA	0.0006	0.1120	0.0134	4.9054	26	1.7500	0.7429
Est Div Yld	0.0010	0.0790	0.0301	5.6644	24	1.7500	0.6857
Est EBIT	0.0009	0.0708	0.0247	7.0005	29	1.7500	0.8286
Est EPS	0.0010	0.0395	0.0302	8.2289	29	1.7500	0.8286
Est Gross Margin	0.0003	0.2315	0.0041	2.3209	19	1.7500	0.5429
Est P/E	0.0010	0.0310	0.0284	7.6941	28	1.7500	0.8000
Est P/Sales	0.0010	0.0506	0.0326	6.0388	27	1.7500	0.7714
Est Price	0.0028	0.0272	0.2026	26.8774	32	1.7500	0.9143
Curr EV	0.0014	0.0079	0.0493	12.0312	34	1.7500	0.9714
FCF	0.0010	0.0441	0.0251	6.8368	28	1.7500	0.8000
Fund Flow	0.0002	0.2170	0.0014	2.0518	18	2.0000	0.4500
Div Yld	0.0008	0.0719	0.0216	3.9415	27	1.7500	0.7714
P/E	0.0013	0.0168	0.0435	10.4918	33	1.7500	0.9429
P/Sales	0.0013	0.0279	0.0482	7.5034	32	1.7500	0.9143
Volume	0.0004	0.0723	0.0038	3.8007	33	2.0000	0.8250

Table 17: Summary of OLS regression results across factors

The results are quite convincing, as statistically significant factors are identified at a rate substantially higher than would be expected by chance. While some of the p -values and R^2 values are relatively weak, the corresponding t -statistics are generally quite strong, providing further evidence of a meaningful relationship.

The adjustment does improve the results, as running the same regressions using the raw data produces a much weaker relationship, as shown in the table below.

variable field	Adjusted				Raw			
	Coeff	<i>p</i> -value	R^2	<i>t</i> -value	Coeff	<i>p</i> -value	R^2	<i>t</i> -value
Curr EV	0.0014	0.0079	0.0493	12.0312	0.0007	0.1621	0.0137	5.5368
Div Yld	0.0008	0.0719	0.0216	3.9415	0.0003	0.2238	0.0035	2.0788
Est Div Yld	0.0010	0.0790	0.0301	5.6644	0.0010	0.0420	0.0234	6.6171
Est EBIT	0.0009	0.0708	0.0247	7.0005	0.0003	0.2081	0.0036	2.5824
Est EPS	0.0010	0.0395	0.0302	8.2289	0.0004	0.1962	0.0042	2.7948
Est EV/EBITDA	0.0006	0.1120	0.0134	4.9054	0.0003	0.1360	0.0028	2.6092
Est FCF	0.0004	0.1863	0.0048	2.6748	0.0002	0.2838	0.0014	1.3826
Est Gross Margin	0.0003	0.2315	0.0041	2.3209	0.0001	0.3727	0.0006	1.0269
Est P/E	0.0010	0.0310	0.0284	7.6941	0.0005	0.2230	0.0078	4.2119
Est P/Sales	0.0010	0.0506	0.0326	6.0388	0.0005	0.1666	0.0087	3.6889
Est Price	0.0028	0.0272	0.2026	26.8774	0.0014	0.0832	0.0532	11.3179
FCF	0.0010	0.0441	0.0251	6.8368	0.0005	0.1856	0.0063	3.2437
Fund Flow	0.0002	0.2170	0.0014	2.0518	0.0003	0.1035	0.0030	3.3824
P/E	0.0013	0.0168	0.0435	10.4918	0.0005	0.1366	0.0075	4.3036
P/Sales	0.0013	0.0279	0.0482	7.5034	0.0004	0.1509	0.0056	3.0868
Volume	0.0004	0.0723	0.0038	3.8007	0.0006	0.0513	0.0084	5.7580

Table 18: Comparing Adjusted vs. Raw Fundamental Factor OLS Results

Similar to prior approaches run the same regression except with it lagged to measure the predictiveness. The quality of the regression doesn't change much in terms of R^2 or the β coefficients. The p -values and t -stats are relatively worse.

Field	No Lag				With Lag			
	Coeff	<i>p</i> -value	R^2	<i>t</i> -value	Coeff	<i>p</i> -value	R^2	<i>t</i> -value
Curr EV	0.0014	0.0079	0.0493	12.0312	0.0003	0.1350	0.0041	2.4920
Div Yld	0.0008	0.0719	0.0216	3.9415	0.0003	0.3527	0.0021	1.2221
Est Div Yld	0.0010	0.0790	0.0301	5.6644	0.0003	0.3758	0.0024	1.4085
Est EBIT	0.0009	0.0708	0.0247	7.0005	0.0002	0.4862	0.0011	1.0183
Est EPS	0.0010	0.0395	0.0302	8.2289	0.0002	0.3964	0.0015	1.2359
Est EV/EBITDA	0.0006	0.1120	0.0134	4.9054	0.0003	0.2500	0.0035	1.9752
Est FCF	0.0004	0.1863	0.0048	2.6748	0.0001	0.5247	0.0003	0.6919
Est Gross Margin	0.0003	0.2315	0.0041	2.3209	0.0002	0.4459	0.0018	0.9400
Est P/E	0.0010	0.0310	0.0284	7.6941	0.0003	0.2746	0.0040	2.1071
Est P/Sales	0.0010	0.0506	0.0326	6.0388	0.0003	0.2788	0.0028	1.5824
Est Price	0.0028	0.0272	0.2026	26.8774	0.0003	0.4642	0.0076	1.8174
FCF	0.0010	0.0441	0.0251	6.8368	0.0003	0.2504	0.0044	1.7336
Fund Flow	0.0002	0.2170	0.0014	2.0518	0.0001	0.3479	0.0004	1.1746
P/E	0.0013	0.0168	0.0435	10.4918	0.0003	0.2415	0.0046	2.0280
P/Sales	0.0013	0.0279	0.0482	7.5034	0.0003	0.3751	0.0027	1.2485
Volume	0.0004	0.0723	0.0038	3.8007	0.0002	0.3399	0.0008	1.5016

Table 19: Comparing Adjusted Instantaneous and Lagged OLS Regression

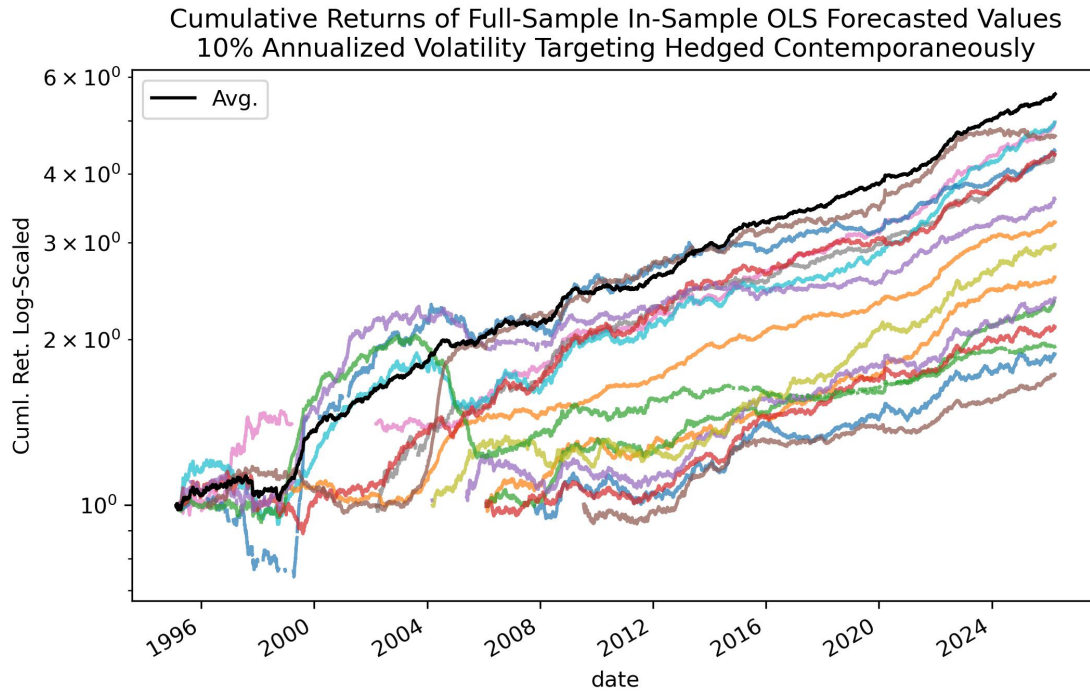


Figure 14: Backtest of Full-Sample/In-Sample OLS Forecasted Returns

The sharpes for the portfolios with the same train/test sample split have the resulting returns.

Field	Expanding	Full-Sample	Train/Test
Avg	0.92	1.99	2.15
Curr EV	0.70	1.17	0.96
Div Yld	0.17	0.81	0.59
Est Div Yld	0.78	1.75	1.35
Est EBIT	0.20	0.96	0.93
Est EPS	0.21	0.97	0.93
Est EV/EBITDA	0.79	1.43	1.40
Est FCF	0.21	0.82	0.79
Est Gross Margin	0.29	1.07	0.54
Est P/E	0.66	1.39	1.35
Est P/Sales	1.00	1.91	1.51
Est Price	0.54	1.21	1.21
FCF	0.74	1.22	1.24
Fund Flow	0.61	1.92	2.01
P/E	0.73	1.03	0.74
P/Sales	0.69	1.26	1.09
Volume	0.77	1.62	1.54

The resulting PnL curve for equal weight across those strategies.

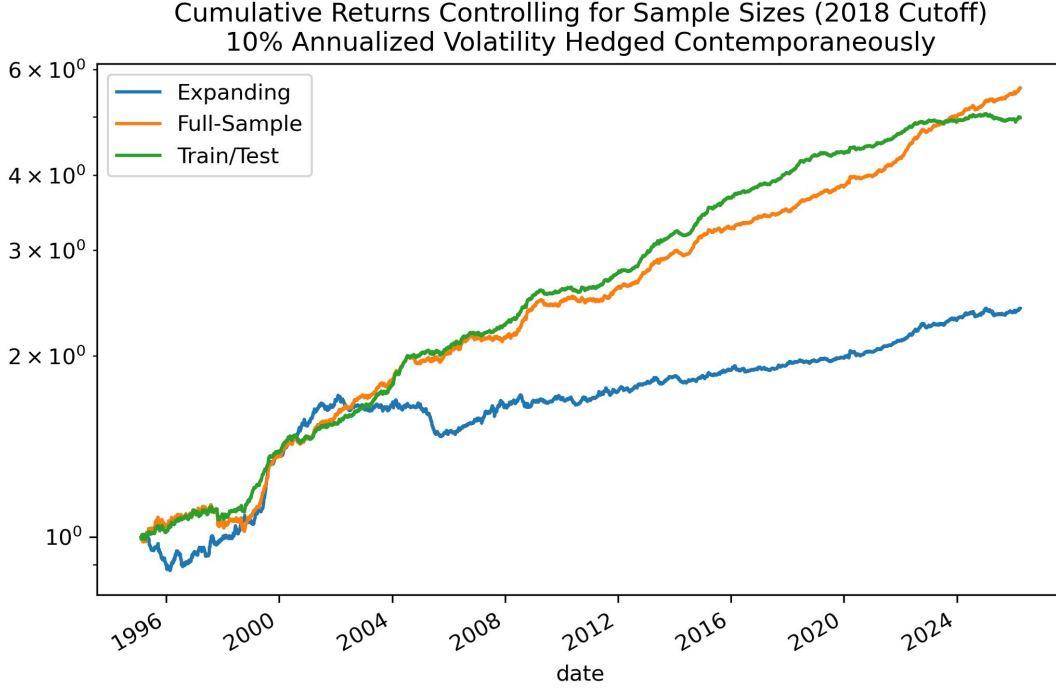


Figure 15: Backtest of equal factor portfolio using OLS forecasted returns

The resulting sharpes of those portfolios are in the table below.

Vol	Lagged	Perfect
Expanding	0.83	0.92
Full-Sample	1.80	1.99
Train/Test	1.94	2.15

8 Machine Learning Factors (Working Section)

The tables from the prior factor model regressions suggest that the framework can potentially be enhanced through machine learning. The overall goal is to expand upon the existing framework by introducing a non-linear approach to finding α , rather than attempting to identify an entirely novel source of non-linear α . As seen in the prior regressions, the factors exhibit relatively strong t -statistics, weak R^2 values, and perform well in the backtest, although the p -values are not particularly strong. Overall, there is evidence that the factors contain predictive information, both within the regression framework and in the backtest. The next step is therefore to use alternative regression approaches to fit these relationships and determine whether a non-linear model can improve upon the linear specification. All the models are fit on volatility targeted FX returns with contemporaneous hedging.

Ridge, Elastic Net, Polynomial, Random Forest, and XGBoost models were chosen as natural extensions beyond linear regression. Expanding out-of-sample and full-sample approaches are either highly susceptible to overfitting or computationally intensive. Therefore, a train/test split with a 2018 cutoff is used. All results are reported using this approach for consistency and future reference.

Some of the model results are somewhat difficult to believe, with the most likely source being overfitting. There is high confidence in the thesis and underlying code, and there is clear empirical evidence of overfitting in the machine learning results. Within the linear model, the in-sample and out-of-sample results were rigorously examined, making the model specifications and hyperparameters the more likely source of the overfitting. The codebase remains under active development to address these issues. The results are nevertheless reported for full transparency and to provide a reference point for further development.

Below is the correlation between each models predicted values and the OLS predicted values. Based on the table it's quite evident that the Ridge and Elastic Net generate values that are quite close to the OLS model. The difference between Ridge and Elastic Net imply that some of the factors get *turned off*.

Group model	In-Sample				Out-Sample			
	EM Carry	Ill Fut	G10 Carry	Liq Fut	EM Carry	Ill Fut	G10 Carry	Liq Fut
Elastic Net	0.77	0.61	NaN	NaN	0.68	0.62	NaN	NaN
Polynomial	0.55	0.38	0.38	0.40	0.46	0.28	0.27	0.33
Random Forest	0.65	0.62	0.52	0.58	0.62	0.58	0.52	0.56
Ridge	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
XGBoost	0.58	0.50	0.47	0.54	0.58	0.49	0.47	0.54

Table 20: Correlation of forecasted values to OLS controlling for in-sample and out-of-sample

Building the backtest using a similar approach, consistent with the methodology applied throughout this paper.

Group Sample Group	EM Carry		G10 Carry		Ill Fut		Liq Fut	
	IS	OS	IS	OS	IS	OS	IS	OS
Elastic Net	0.99	2.59	0.46	0.81	0.61	-0.03	0.62	0.37
Linear	2.40	3.47	2.03	0.77	1.32	0.86	2.32	0.83
Polynomial	4.39	3.14	5.92	0.54	3.29	0.69	6.61	0.86
Random Forest	4.35	3.25	4.06	0.50	2.94	0.79	4.44	1.08
Ridge	2.39	3.49	2.03	0.77	1.32	0.84	2.33	0.82
XGBoost	6.18	3.16	6.04	-0.18	4.09	0.82	6.95	0.70

Table 21: Sharpes for backtested models

The equal weight portfolio has the following sharpes.

Vol. Target Model	Lagged		Perfect	
	In-Sample	Out-Sample	In-Sample	Out-Sample
Elastic Net	1.25	2.24	1.41	2.40
Linear	3.15	2.46	3.31	2.50
Polynomial	7.61	2.45	7.81	2.50
Random Forest	6.00	2.42	6.24	2.47
Ridge	3.15	2.45	3.31	2.50
XGBoost	8.72	2.01	9.03	2.07

Table 22: Sharpes of Equal Weight Portfolio Controlling for In-Sample and Out-Sample

The out-of-sample performance for the strategies are plotted on the next page.

Cumulative Returns of Out-of-Sample Returns by Model
10% Annualized Volatility Targeting Hedged Contemporaneously

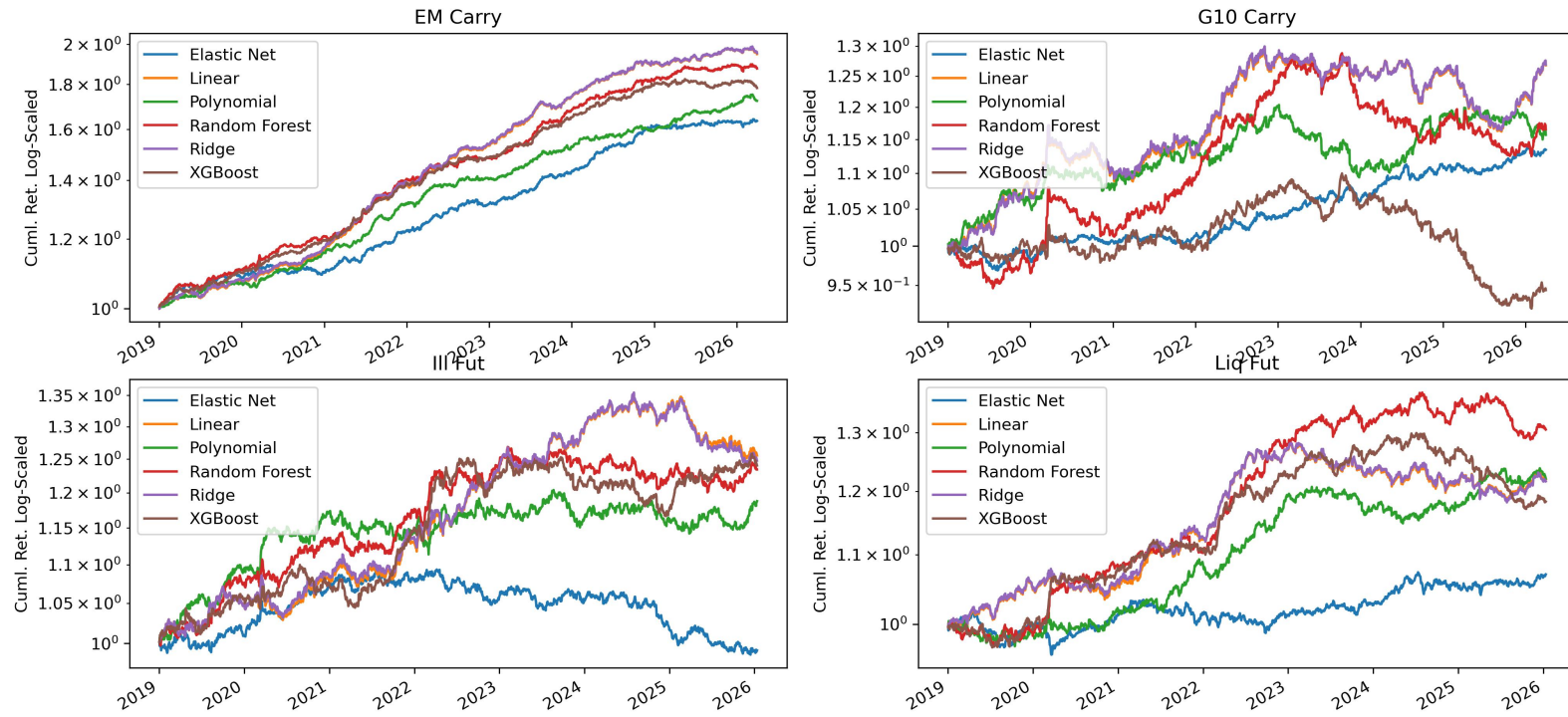


Figure 16: Backtest of out-of-sample ML Models

Although the out-of-sample performance remains relatively strong, there is a substantial reduction in Sharpe when moving out of sample, which suggests that the model is overfitting.

	Lagged	Perfect
Model		
Elastic Net	1.00	0.99
Linear	-0.70	-0.80
Polynomial	-5.16	-5.31
Random Forest	-3.58	-3.77
Ridge	-0.70	-0.81
XGBoost	-6.71	-6.95

Table 23: In-Sample - Out-Sample Sharpe

The last part is to quantify how much the machine learning models are contributing. A useful way to examine this is to compare the difference in RMSE with Sharpe performance. The in-sample relationship is quite clear: as the RMSE spread becomes smaller, the Sharpe performance improves substantially. This relationship is consistent across model choices, although it does not hold up out of sample.

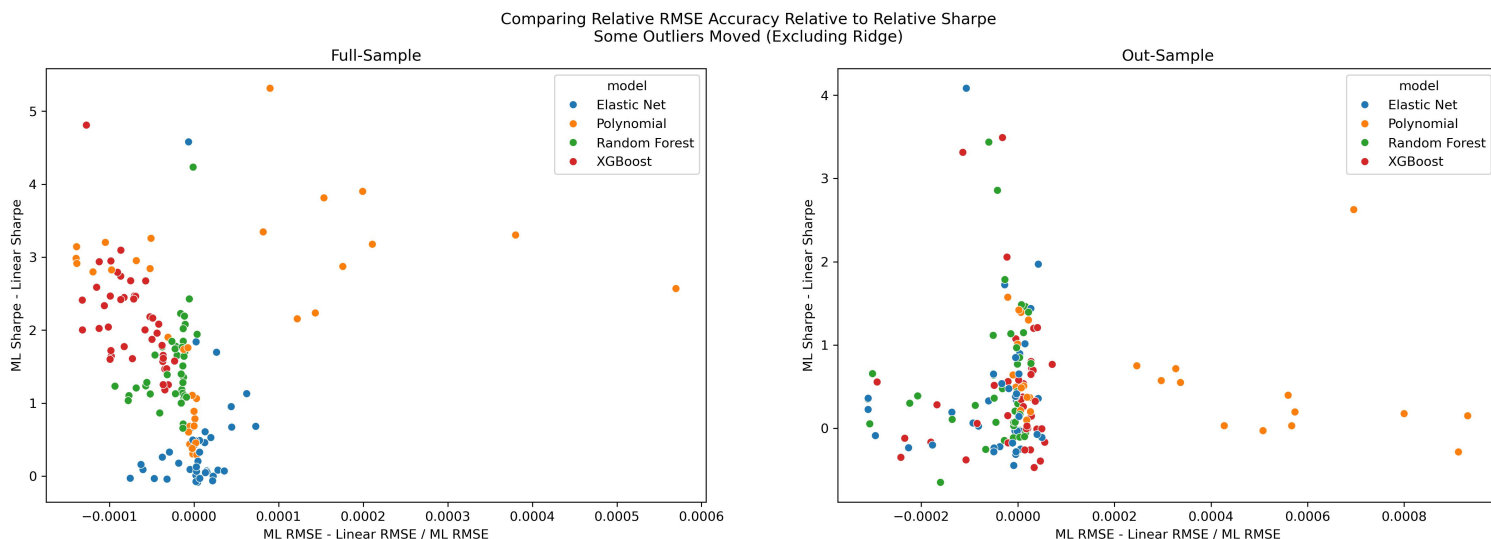


Figure 17: Comparison of difference between RMSE and Sharpe controlling for in-sample and out-sample

9 Single Name Stock Approach (Working)

Another extension is to use the individual stocks comprising each ETF. The stock-level data was collected from Yahoo Finance, while the ETF holdings were obtained directly from the respective companies. It should be noted that the stock data does not account for survivorship bias or changes in index composition. However, many of the usual concerns associated with these issues are likely less relevant in this case, as the stock returns are not directly being used for trading.

Below is a table showing the number of securities that correspond to each ticker.

	ETF Ticker	Number of Stocks	FX Ticker
16	EWT	7	USDTWDCR Curncy
15	EWS	11	USDSGDCR Curncy
8	EPU	11	USDPENCR Curncy
0	COLO	16	USDCOPCR Curncy
12	EWB	18	USDHKDCR Curncy
21	EZA	22	USDZARCR Curncy
1	ECH	22	USDCLPCR Curncy
6	ENZL	24	USDNZDCR Curncy-NV1 Curncy
7	EPOL	25	USDPLNCR Curncy-PP1 Curncy
18	EWV	26	PE1 Curncy-USDMXNCR Curncy
11	EWD	34	USDSEKCR Curncy-SE1 Curncy
2	EDEN	35	USDDKKCR Curncy
20	EWZ	35	BR1 Curncy-USDBRLCR Curncy
14	EWL	36	SF1 Curncy-USDCFPCR Curncy
5	ENOR	40	NO1 Curncy-USDNOKCR Curncy
3	EIDO	42	USDIDRCR Curncy
9	EWA	43	USDAUDCR Curncy-AD1 Curncy
17	EWU	60	BP1 Curncy-USDGBPCR Curncy
19	EWY	62	USDKRWCR Curncy-KO1 Curncy
25	THD	62	USDTHBCR Curncy
10	EWG	75	CD1 Curncy-USDCADCR Curncy
4	EIS	87	IS1 Curncy
23	INDA	130	USDINRCR Curncy-SIR1 Curncy
13	EWJ	161	USDJPYCR Curncy-JY1 Curncy
22	EZU	189	USDEURCR Curncy-EC1 Curncy
24	MCHI	368	UCA1 Curncy

To keep things simple, start with running a linear regression using single stocks vs ETF returns. SPY returns are subtracted from the ETF and single name returns and then z-scored for the momentum signal. The first step is to compare the additive R^2 and β which is quite small⁴

Lag	Lagged		No Lag	
	β	R^2	β	R^2
Group				
EM Carry	0.0004	0.0143	0.0026	-0.0848
G10 Carry	-0.0001	0.0351	0.0030	-0.0391
Ill Fut	-0.0001	0.0421	-0.0027	-0.1037
Liq Fut	-0.0001	0.0315	-0.0031	-0.0466

The same OLS regression is then run to predict futures returns, and the Sharpe ratio of the resulting portfolio is shown below.

Model Sample Group Group	Full-Sample Full-Sample	Train/Test	
		In-Sample	Out-Sample
EM Carry	2.99	3.04	0.99
G10 Carry	1.85	2.01	-0.82
Ill Fut	3.38	3.67	0.19
Liq Fut	2.56	2.80	0.16

⁴Although it was already small to begin with from the prior regressions

Now that the baseline results have been established the next step is to compare how much the laged volatility target contributes.

regression variable group	Full-Sample		Train/Test	
	Lagged	Perfect	Lagged	Perfect
EM Carry	2.65	2.99	2.38	2.63
G10 Carry	1.79	1.85	1.47	1.56
Ill Fut	3.28	3.38	2.71	2.77
Liq Fut	2.48	2.56	2.20	2.29

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